Optimization

Compiler Sessional Online

Problem Statement

Add a command called `A` for assignment operation. The syntax goes like this -

```
A foo
```

This assigns a **constant** value (such as 2, 3 etc.) to the variable `foo`.

Assumption

You can safely assume that the variable already exists in the symbol table. No need to check for existence or handle errors regarding non-existence. You do not need to worry about the type of the variable as well.

Now implement the following -

- Upon exiting a scope, report all the **unused variables**
- Report all the **Dead Stores**

Unused Variable

A variable is considered unused if it **was never looked up**.

Dead Store

A dead store is an assignment operation that is not used or read. For example -

```
A foo INT
...
...
A foo INT
...
...
E
```

If there was no lookups for `foo` made in between the 2 assignments (store operations), then the first one is a dead store.
Also, if no lookups were made after the second assignment and before exiting, then the second assignment is also a dead store.
To summarize, an assignment can be called a valid one (and **not** a dead store) if there is a lookup made after the assignment and before the next assignment or scope exit.

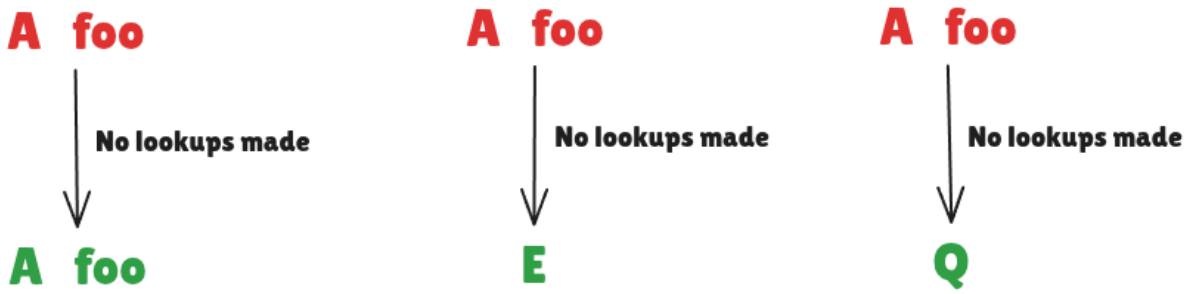


Figure 1: In all 3 cases, the assignment in red is a dead store

Restrictions

- You **can** use STL
- You **can** modify existing classes from your practice assignment
- Must compile with `-fsanitize=address`

Sample input, output, and explanations are on the following pages.

Examples

Sample Input

```
3
I a VAR
S
I a VAR
A a
I b VAR
A a
L a
A b
E
A a
Q
```

Sample Output

```
 ScopeTable# 1 created
Cmd 1: I a VAR
 Inserted in ScopeTable# 1 at position 2, 1
Cmd 2: S
 ScopeTable# 2 created
Cmd 3: I a VAR
 Inserted in ScopeTable# 2 at position 2, 1
Cmd 4: A a
 Assigned 'a' in ScopeTable# 2
Cmd 5: I b VAR
 Inserted in ScopeTable# 2 at position 3, 1
Cmd 6: A a
 Dead store: Cmd 4: A a
 Assigned 'a' in ScopeTable# 2
Cmd 7: L a
 'a' found in ScopeTable# 2 at position 2, 1
Cmd 8: A b
 Assigned 'b' in ScopeTable# 2
Cmd 9: E
 Dead store: Cmd 8: A b
 Unused variable: 'b'
 ScopeTable# 2 removed
Cmd 10: A a
 Assigned 'a' in ScopeTable# 1
Cmd 11: Q
 Dead store: Cmd 10: A a
 Unused variable: 'a'
 ScopeTable# 1 removed
```

Explanation

1. The assignment in command 4 is a dead store because there was no lookup made before the next assignment (command 6)
2. The assignment in command 6 is **not** a dead store because there was a lookup made in command 7 before exiting (command 9)
3. The assignment in command 8 is a dead store because there was no lookup made before exiting the scope (command 9)
4. The variables `b` (from the inner scope) and `a` (from the outer scope) are unused variables because they were never looked up
5. The assignment in command 10 is a dead store because there was no lookup made before quitting (command 11)

Marks Distribution

Task	Marks
Report of unused variables	4
Report of dead stores	6
Total	**10**